

Independent Computational Verification

Paper II — Section 7: exact integer maximization (Prop 7.1, Theorem 1.1 arithmetic)

Author of record: Juan Pablo Traverso Gianini Date: 2026-07-09

Artifact: E_integer_maximization/audit_E_integer_max.py Verdict: PASS

Scope

Every integer n from 1 to 2000, in **exact rational arithmetic** (fractions).

Claims verified

- **(E1) Theorem 1.1 value:** $\max_{p+q=n} \Phi_\tau(S_{p,q}) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor$.
- **(E2) maximizer:** the maximum is attained in the **saturated** branch ($q \geq p - 1$) by the integer p nearest to $(2n + 1)/6$.
- **(E3):** the **non-saturated** branch never exceeds the floor.
- **(E4) residue identity:** $(2n + 1)^2 \equiv 1 \text{ or } 9 \pmod{24}$, matching the closed floor.

Results

- E1 violations: **0**
- E2 violations: **0**
- E3 violations: **0**
- E4 violations: **0**
- **Total violations over all $n \leq 2000$: 0.** Runtime 19.9 s.

n	$\lfloor (2n + 1)^2/24 \rfloor$	maximizer p^*
1	0	1
2	1	1
3	2	1
5	5	2
8	12	3
12	26	4
13	30	5
50	425	17
100	1683	34
1000	166833	334

Verdict

PASS. The closed-form maximum, the location of the maximizer in the saturated branch, the domination of the non-saturated branch, and the modular identity all hold for every $n \leq 2000$ with no exception.

*This document reports an **independent computational verification** of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.*